1 Training pairs with exact footprints



both classes see smoothing **41,512 frames**

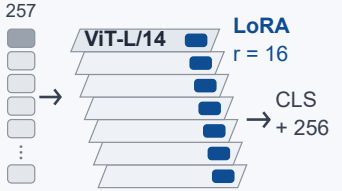


one render → 4 operations **50,112 items**



part-level: outside **M** untouched **61,510 edits**

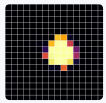
2 One backbone, four heads



forged f, beautified b
P(forged) = σ(f), the score never reads b

4 operations eye, contour, tone, texture

4 measured amounts, auxiliary



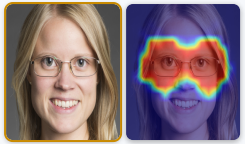
evidence 16×16, vs **M**

3.2 M of 306 M parameters trained

3 Outputs on held-out images



forged 1.00
evidence:
nose



beautified 0.98
evidence:
eyes
unseen service

The model's decision is fake (score 1.00). The evidence is concentrated on the nose (mean evidence 0.99). The score is the model's raw output, not a calibrated probability.

composed by code, checked

4 Three tests that could have failed

Exposure

54% → 12%

smoothed genuine faces accused: no smoothing in training vs **smoothing seen by both classes**

Restoration

94% vs 45–57%

forgery decisions removed by restoring the marked region: **our evidence head** vs heat maps

Matching

51–53% ≈ coin flip

a judge pairs a VLM description with its own photo; so **code writes every sentence**

5 On the device

Browser, iPhone

0.59 s per photo

landmarks, model, part and sentences run in the page; **nothing is uploaded**

genuine · beautified · forged · exact footprint M · trained (LoRA, heads) · frozen